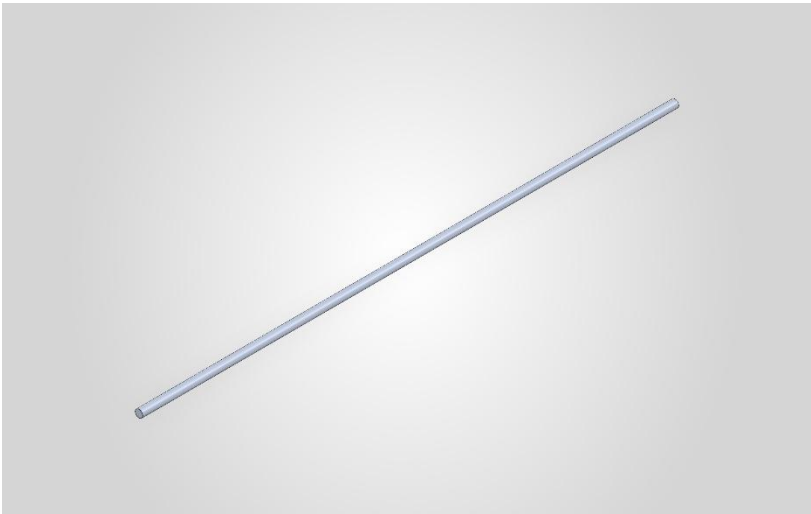




Description
No Data

Simulation of A3 SolidWorks bar

Date: Wednesday, September 9, 2026
Designer: Solidworks
Study name: SimulationXpress Study
Analysis type: Static

Table of Contents

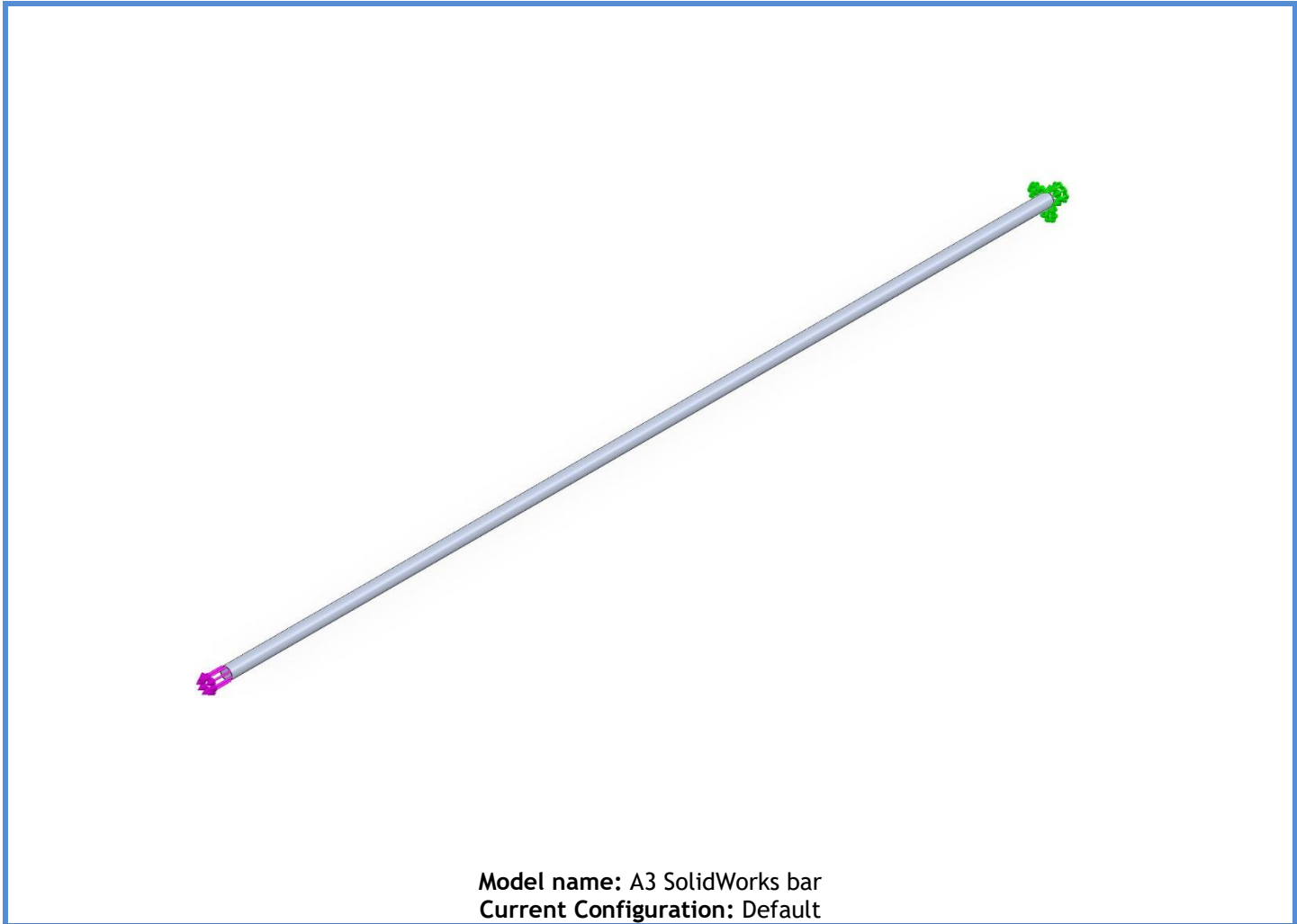
Description 1
Assumptions 2
Model Information..... 3
Material Properties 4
Loads and Fixtures 4
Mesh information..... 5
Study Results..... 7
Conclusion 10



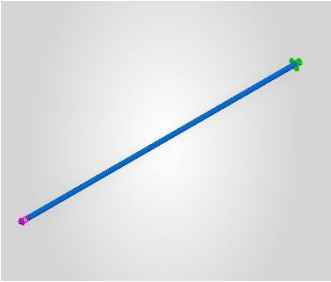
Assumptions



Model Information

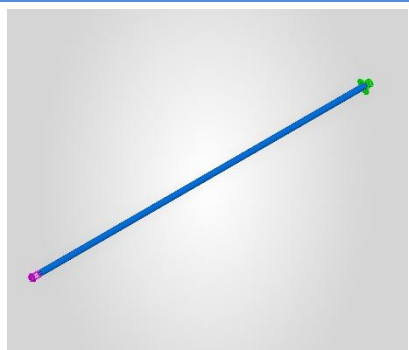


Model name: A3 SolidWorks bar
Current Configuration: Default

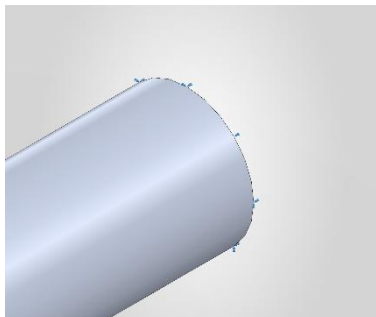
Solid Bodies			
Document Name and Reference	Treated As	Volumetric Properties	Document Path/Date Modified
Boss-Extrude1 	Solid Body	Mass:0.677427 lb Volume:6.94486 in^3 Density:0.0975437 lb/in^3 Weight:0.676968 lbf	C:\Users\jahid\OneDrive\Desktop\MEGR 2157\A3\A3 SolidWorks bar.SLDPRT Sep 9 21:15:55 2026

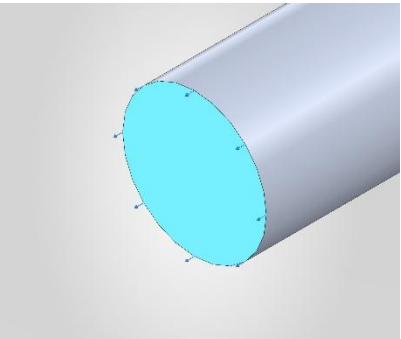


Material Properties

Model Reference	Properties	Components
	Name: 1060 Alloy Model type: Linear Elastic Isotropic Default failure criterion: Max von Mises Stress Yield strength: 3,999.3 psi Tensile strength: 9,998.26 psi	SolidBody 1(Boss-Extrude1)(A3 SolidWorks bar)

Loads and Fixtures

Fixture name	Fixture Image	Fixture Details
Fixed-1		Entities: 1 face(s) Type: Fixed Geometry

Load name	Load Image	Load Details
Force-1 (:Per item: -500 lbf:)		Entities: 1 face(s) Type: Apply normal force Value: -500 lbf



Mesh information

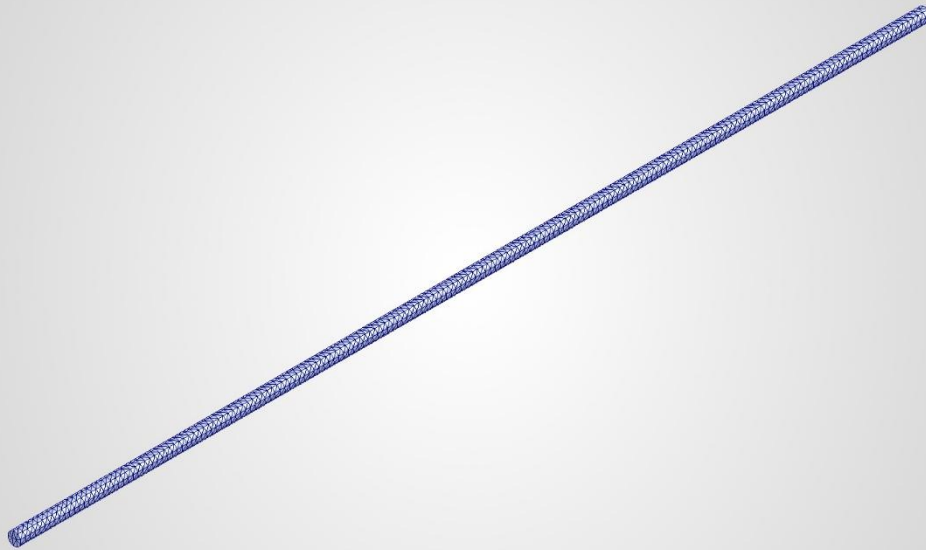
Mesh type	Solid Mesh
Mesher Used:	Standard mesh
Automatic Transition:	Off
Include Mesh Auto Loops:	Off
Jacobian points for High quality mesh	16 Points
Element Size	0.190847 in
Tolerance	0.00954236 in
Mesh Quality	High

Mesh information - Details

Total Nodes	13952
Total Elements	7686
Maximum Aspect Ratio	3.1303
% of elements with Aspect Ratio < 3	100
Percentage of elements with Aspect Ratio > 10	0
Percentage of distorted elements	0
Time to complete mesh(hh:mm:ss):	00:00:01
Computer name:	



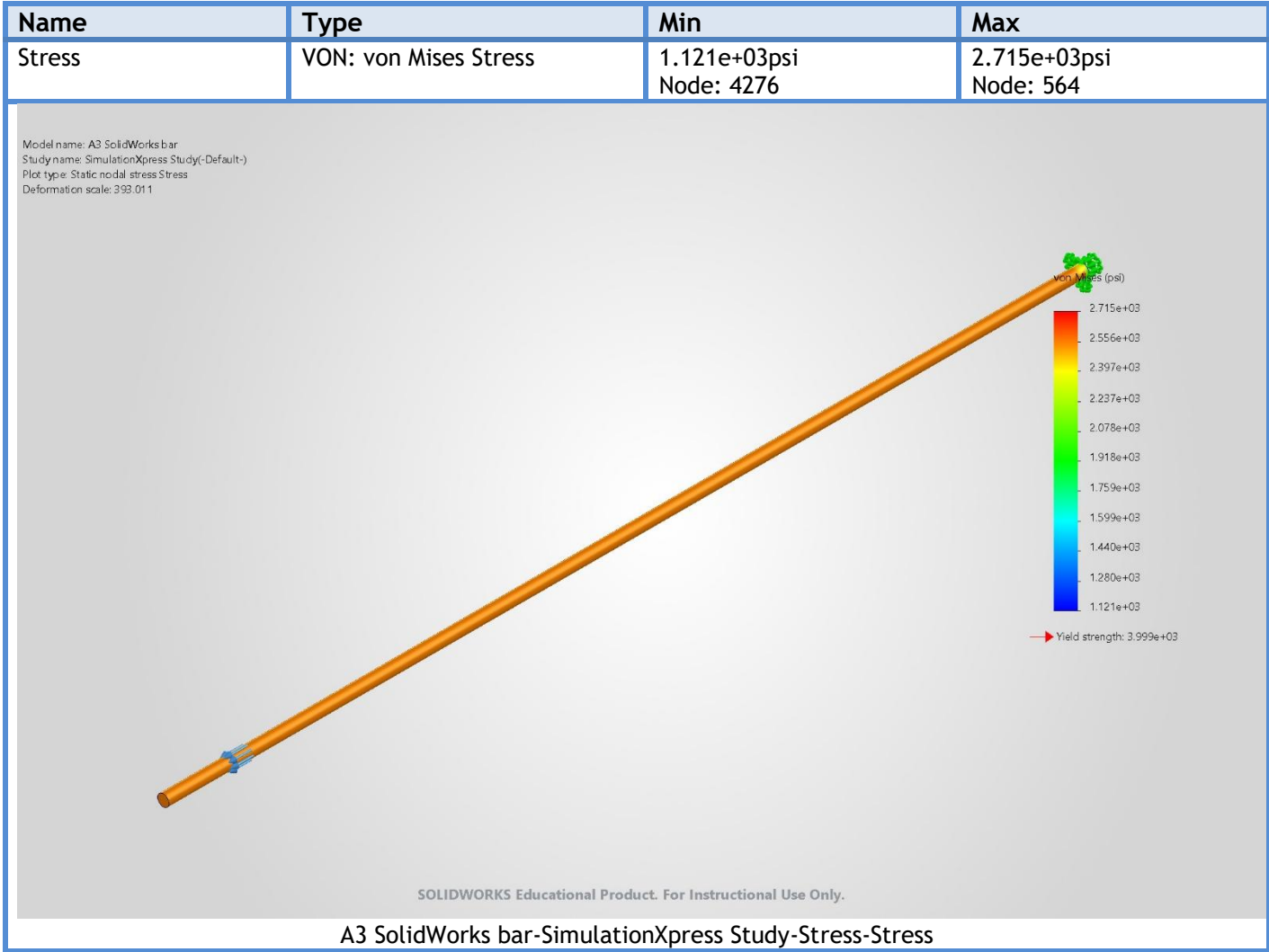
Model name: A3 SolidWorks bar
Study name: SimulationXpress Study(-Default-)
Mesh type: Solid Mesh



SOLIDWORKS Educational Product. For Instructional Use Only.



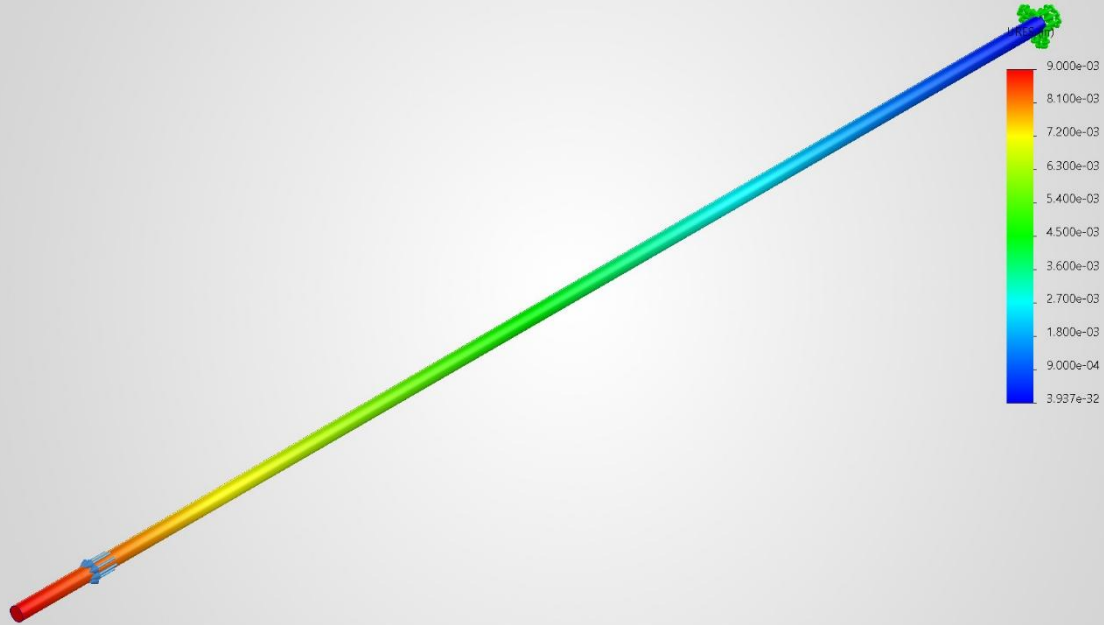
Study Results



Name	Type	Min	Max
Displacement	URES: Resultant Displacement	0.000e+00in Node: 190	9.000e-03in Node: 13946



Model name: A3 SolidWorks bar
 Study name: SimulationXpress Study(-Default-)
 Plot type: Static displacement Displacement
 Deformation scale: 393.011



SOLIDWORKS Educational Product. For Instructional Use Only.

A3 SolidWorks bar-SimulationXpress Study-Displacement-Displacement

Name	Type
Deformation	Deformed shape

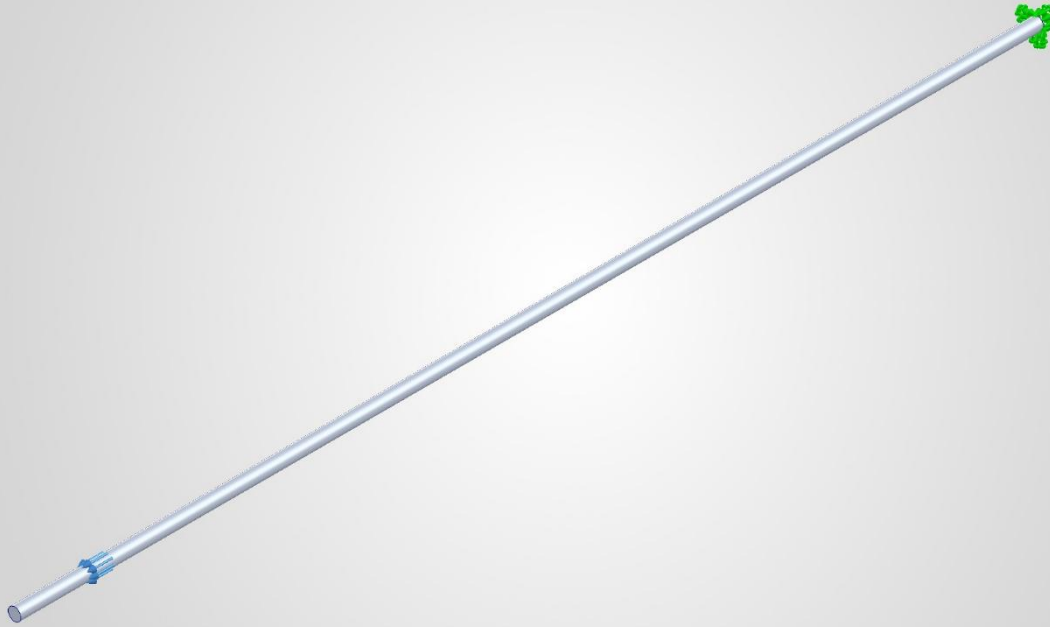


SOLIDWORKS

Analyzed with SOLIDWORKS Simulation

Simulation of A3 SolidWorks bar

Model name: A3 SolidWorks bar
Study name: SimulationXpress Study(-Default-)
Plot type: Deformed shape Deformation
Deformation scale: 393.011



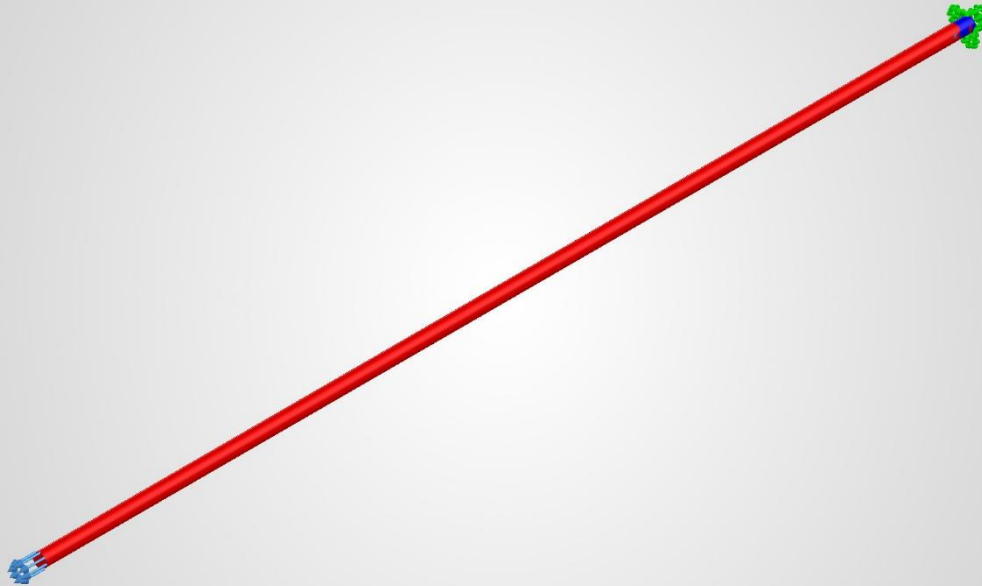
SOLIDWORKS Educational Product. For Instructional Use Only.

A3 SolidWorks bar-SimulationXpress Study-Displacement-Deformation

Name	Type	Min	Max
Factor of Safety	Max von Mises Stress	1.473e+00 Node: 564	3.568e+00 Node: 4276



Model name: A3 SolidWorks bar
Study name: SimulationXpress Study(-Default-)
Plot type: Factor of Safety Factor of Safety
Criterion: Max von Mises Stress
Red < FOS = 1.57 < Blue



SOLIDWORKS Educational Product. For Instructional Use Only.

A3 SolidWorks bar-SimulationXpress Study-Factor of Safety-Factor of Safety

Conclusion

